



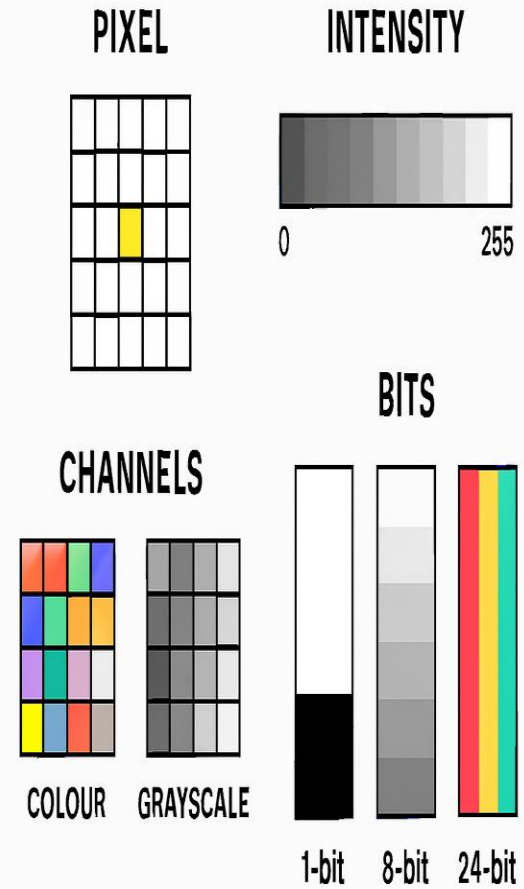
**CSS3913**

**Computer Vision and Image Processing**

**Topic 2**  
**Digital Image**

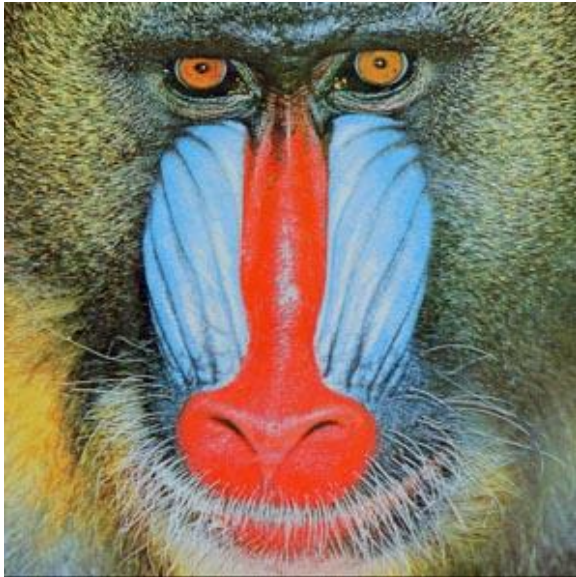
# Some definitions

- **Pixel:** “Picture element” is the smallest unit of a digital image. Each pixel holds color or intensity information.
- **Intensity:** The brightness level of a pixel.
  - Grayscale: Intensity ranges from 0 (black) to 255 (white) in 8-bit images.
  - Color: Intensity is split across 3 channels (e.g., R, G, B), each contributing to the final color.
- **Channels:** Independent layers of color or intensity information.
  - Grayscale: 1 channel.
  - RGB (Colour): 3 channels (Red, Green, Blue).
  - RGBA: 4 channels (adds Alpha for transparency).
  - Multispectral: More than 3 channels (used in remote sensing, medical imaging).
- **Bits:** The number of bits per pixel determines how many intensity or color levels an image can represent.
  - 1-bit: Binary image (black or white).
  - 8-bit: 256 levels (common for grayscale).
  - 24-bit: True color image (8 bits per channel  $\times$  3 channels = 16.7 million colors).



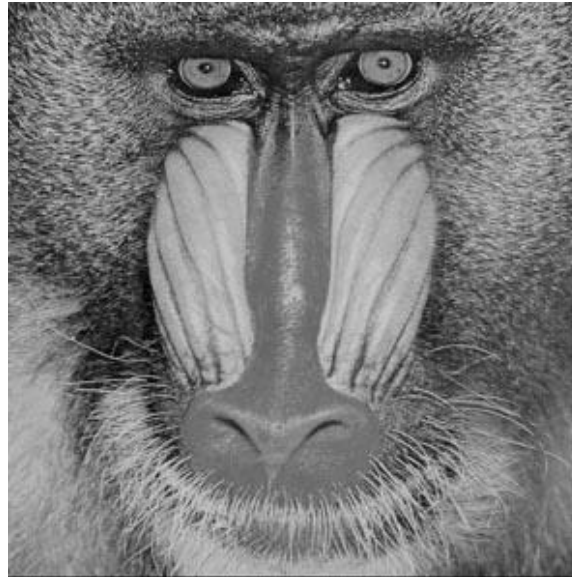
# Type of Images

Colour image



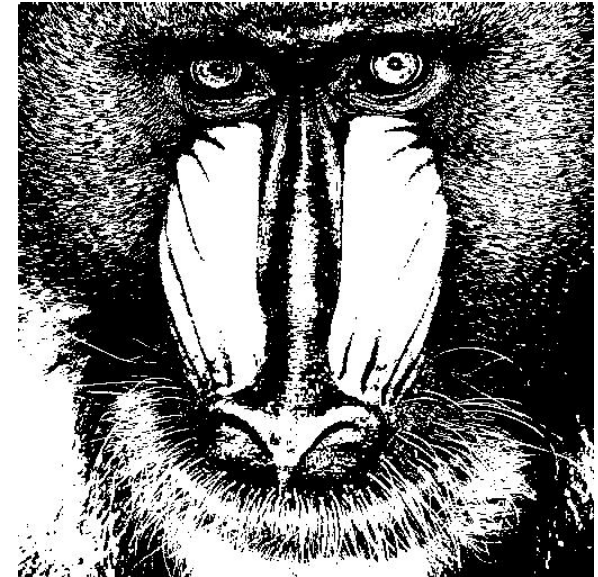
Red channel : 8 bits  
Green channel : 8 bits  
Blue channel : 8 bits  
Intensity levels : 0 ~ 255  
(each channel)

Grayscale image



Single channel : 8 bits  
Intensity levels : 0 ~ 255

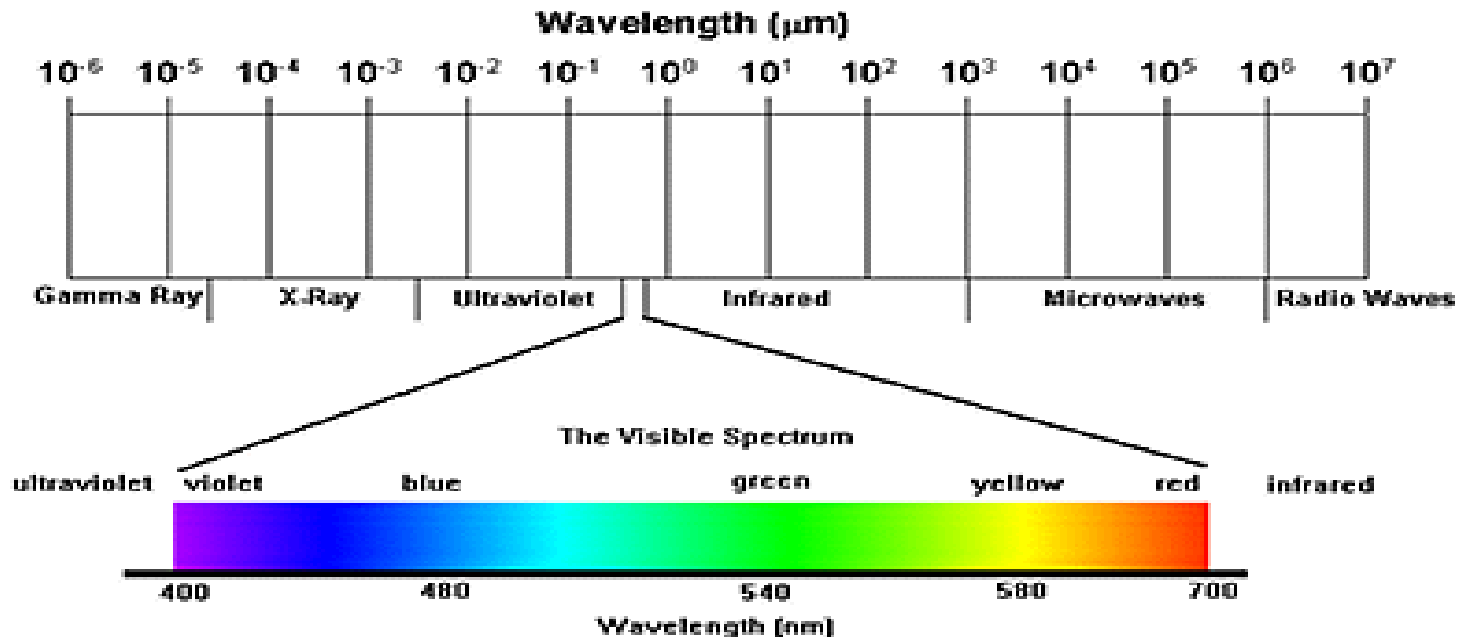
Binary image



Single channel : 1 bits  
Intensity levels : 0 ~ 1

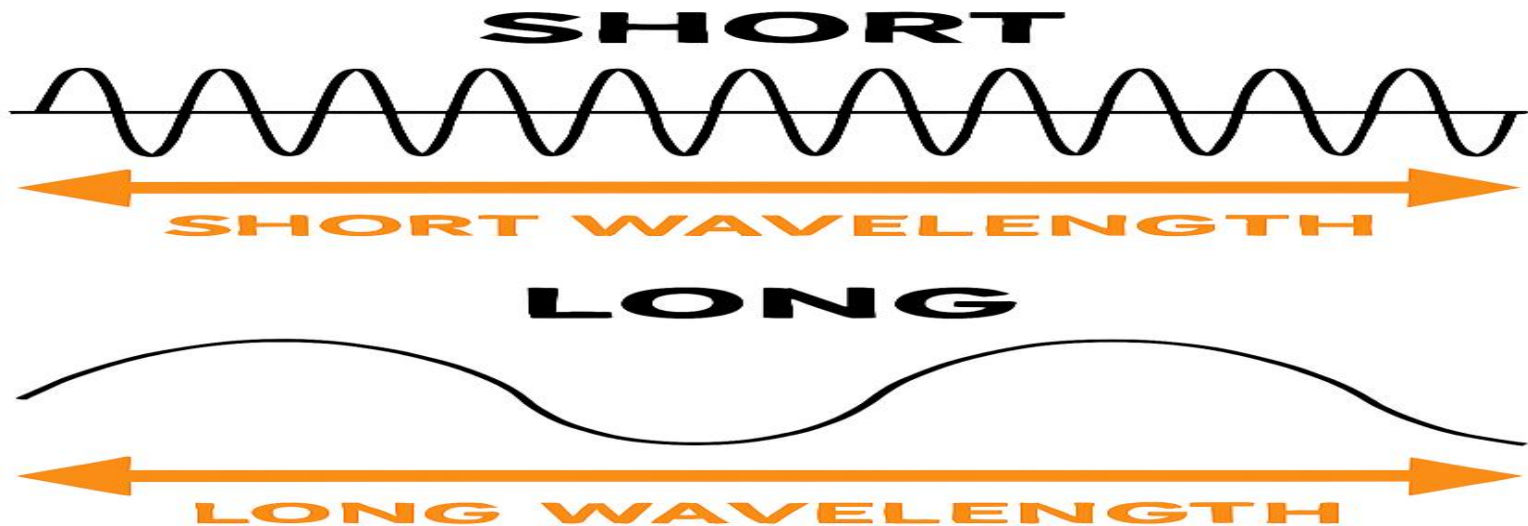
# Sensing light

- ▶ Different kind of radiation that reflect from/ penetrate objects in the physical world can be sensed by different imaging devices.
- ▶ Human eye **sensitive** to radiation with wavelength ranging from 400nm (**violet**) to 700nm (**red**).



# Sensing light (Cont')

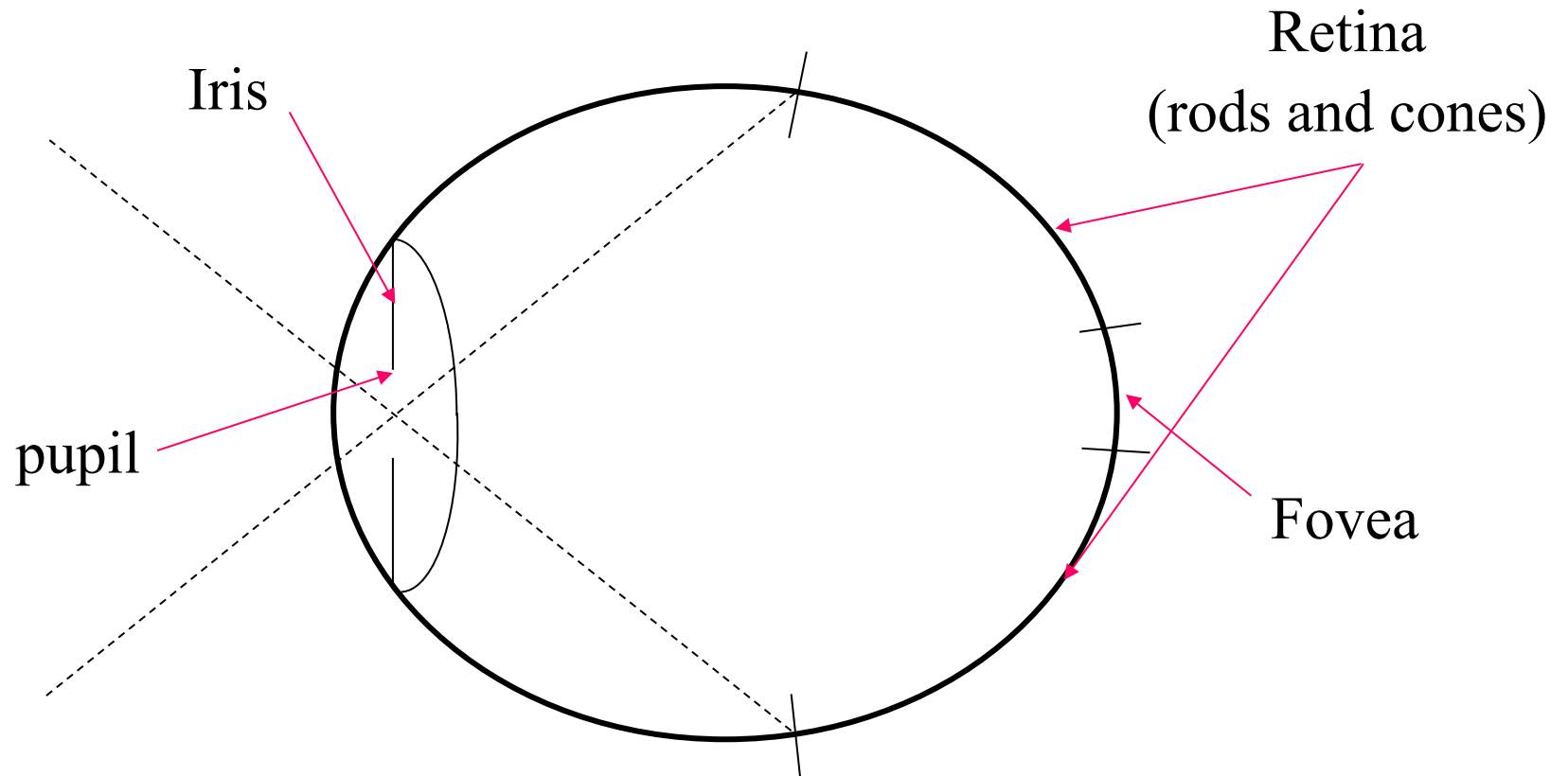
- ▶ Snakes and Charge-Coupled Device sensors (CCD): radiation longer than 800nm (infrared)
- ▶ Different wavelengths of radiation have different properties: x-rays (short) can penetrate bone, while longer wavelengths (infrared) can't even penetrate clouds



# The Human Eye

- The human eye is like a spherical camera with a 20mm focal length lens at the outside focusing the image on the retina which is opposite the lens and fixed on the inside of the surface of the sphere.
  - IRIS – controls the amount of light passing through the lens by controlling the size of the pupil.
  - Each eye has 100 million receptors.
  - The retina is unevenly populated with sensor cells.
  - An area near the centre (fovea) has a very dense population of colour receptors called cones.
  - Away from the center, cones decrease and rods (black-white receptors) increase.
- 

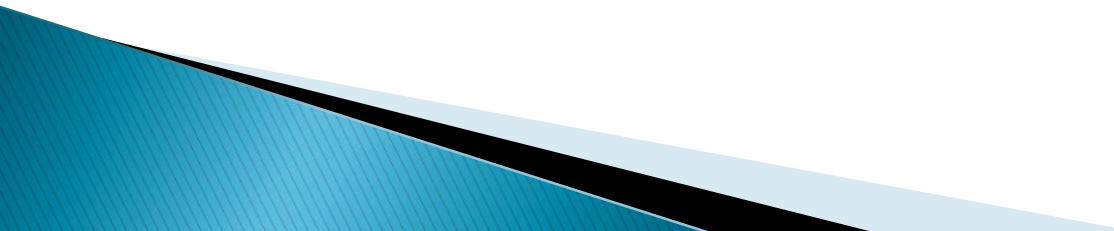
# The Human Eye (cont')



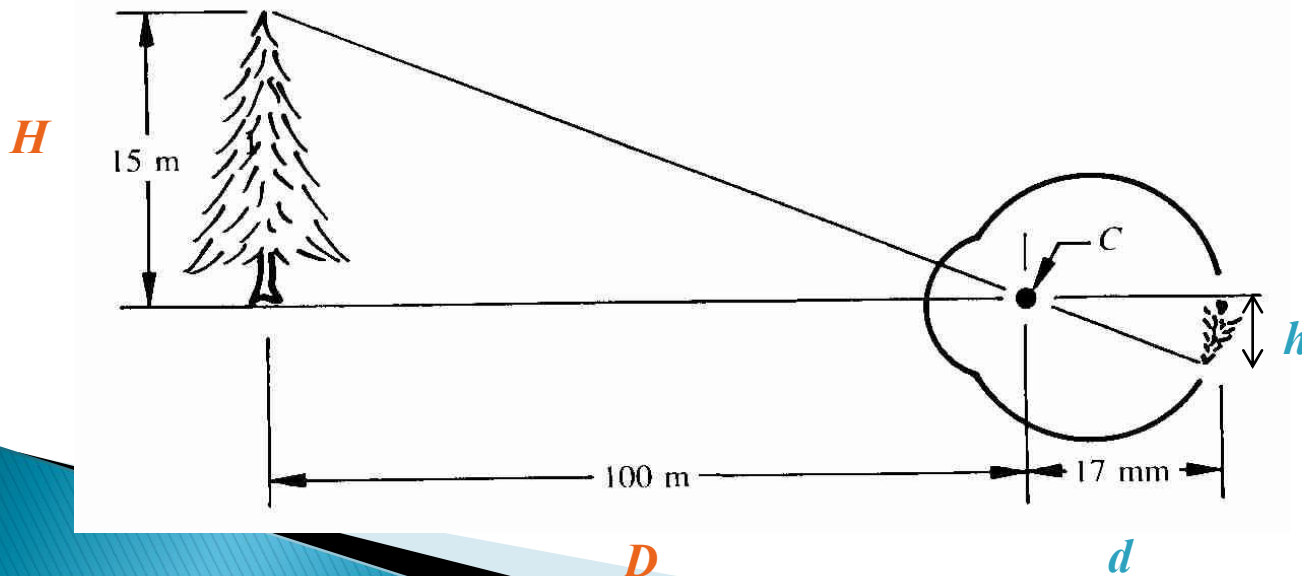
# Image Formation in The Eye

- The principal difference between the lens of the eye and an ordinary optical lens is that the former is flexible.
  - The radius of curvature of the anterior surface of the lens is greater than the radius of its posterior surface.
  - The shape of the lens is controlled by the tension in the fibers of the ciliary body.
  - To focus on distant objects, the controlling muscles cause the lens to be relatively flattened.
  - Similarly, these muscles allow the lens to become thicker in order to focus on objects near the eye.
- 

# Image Formation in The Eye (Cont')

- The distance between the focal center of the lens and the retina varies from approximately 17 mm to about 14 mm, as the refractive power of the lens increases from its minimum to its maximum.
  - When the eye focuses on an object farther away than about 3 m, the lens exhibits its lowest refractive power, and when the eye focuses on a nearby object the lens is most strongly refractive.
  - This information makes calculating the size of the retinal image of any object easy.
- 

- E.g. The observer is looking at a tree 15 m high at a distance of 100 m. If  $h$  is the size of the retinal image in millimeters, the geometry of the Figure below yields  $15\text{m}/100\text{m} = h/17\text{mm}$  or  $h = 2.55\text{ mm}$ . The retinal image is reflected primarily in the area of the fovea. Perception then occurs through the relative excitation of light receptors, which transform radiant energy into electrical impulses that are ultimately decoded by the brain.



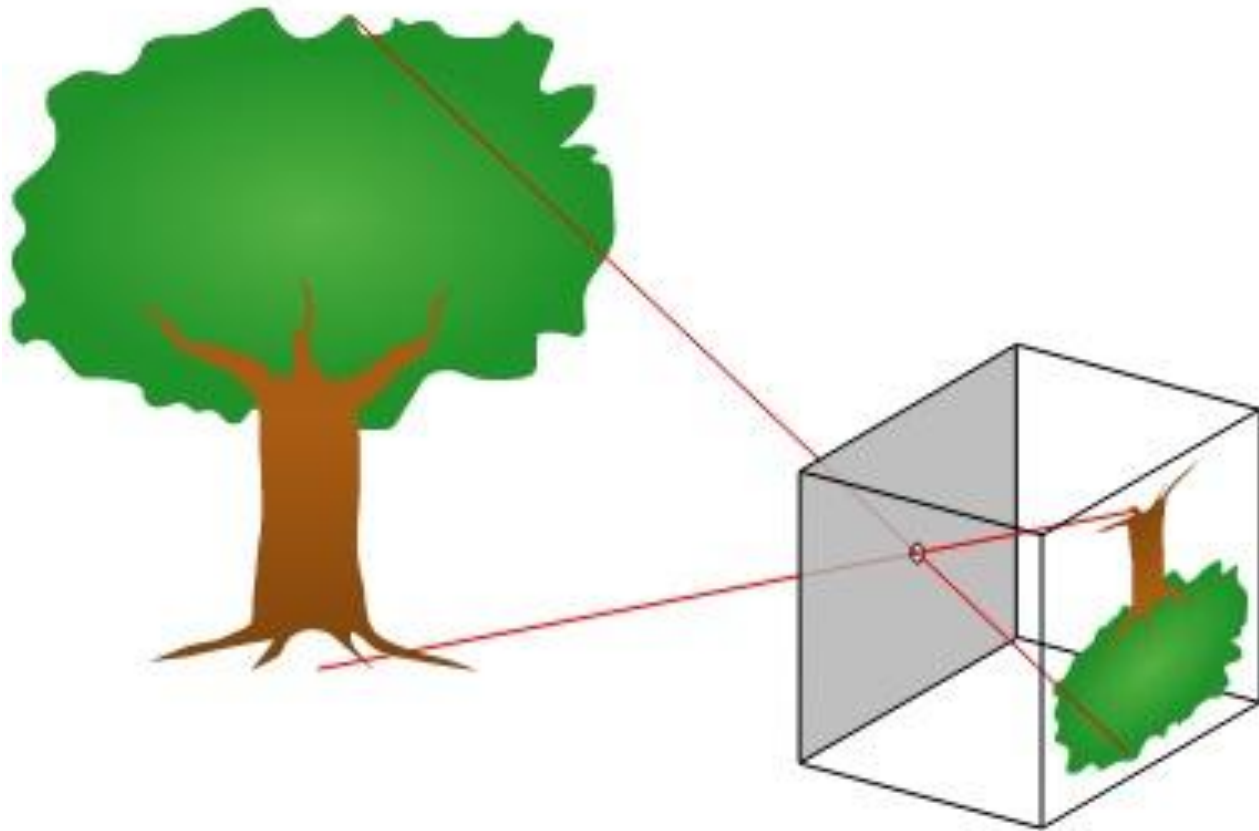
$$H / D = h / d$$

# Image Geometry

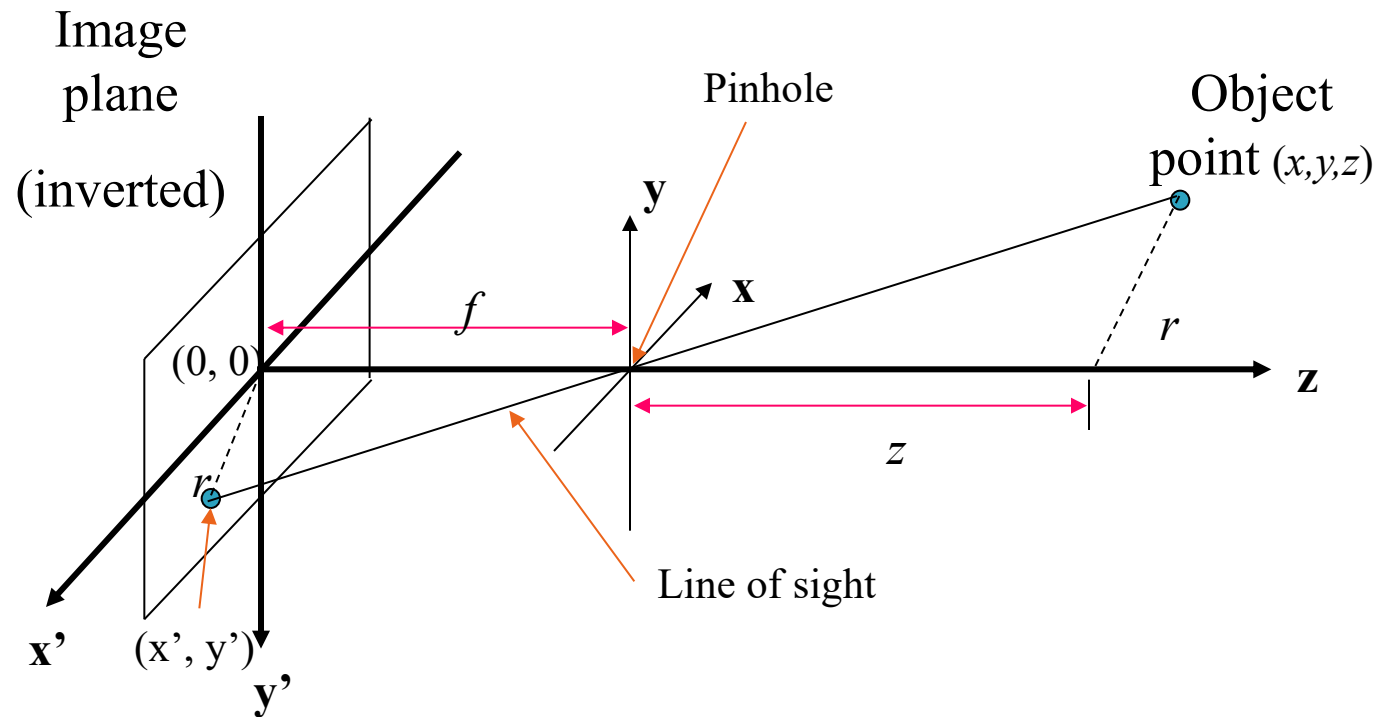
- ▶ There are two parts to the image formation process:
  - 1) The **geometry** of image formation, which determines where in the image plane the projection of a point in the scene will be located.
  - 2) The **physics of light**, which determines the brightness of a point in the image plane as a function of scene illumination and surface properties.
- ▶ The basic model for the projection of points in the scene onto the image plane (**3-D** to **2-D** mapping) is diagrammed in the Figure next page known as the *pinhole model*.
- ▶ In the pinhole model, the imaging system's centre of projection coincides with the origin of the 3-D coordinate system.

# Image Geometry (Cont')

## Pinhole model

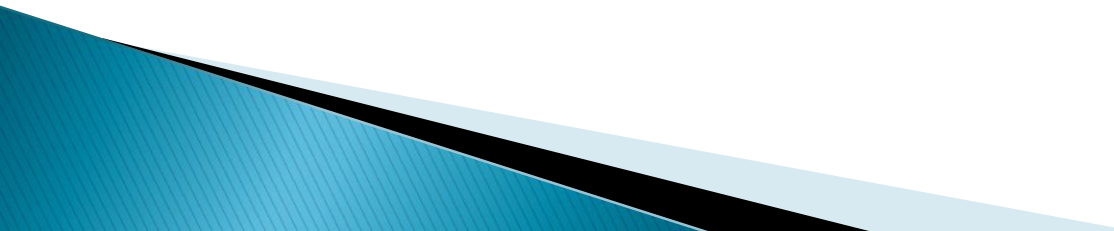


# Image Geometry (cont')



The pinhole model. The point on the image plane that corresponds to a particular point in the scene is found by following the line that passes through the scene and the center of projection.

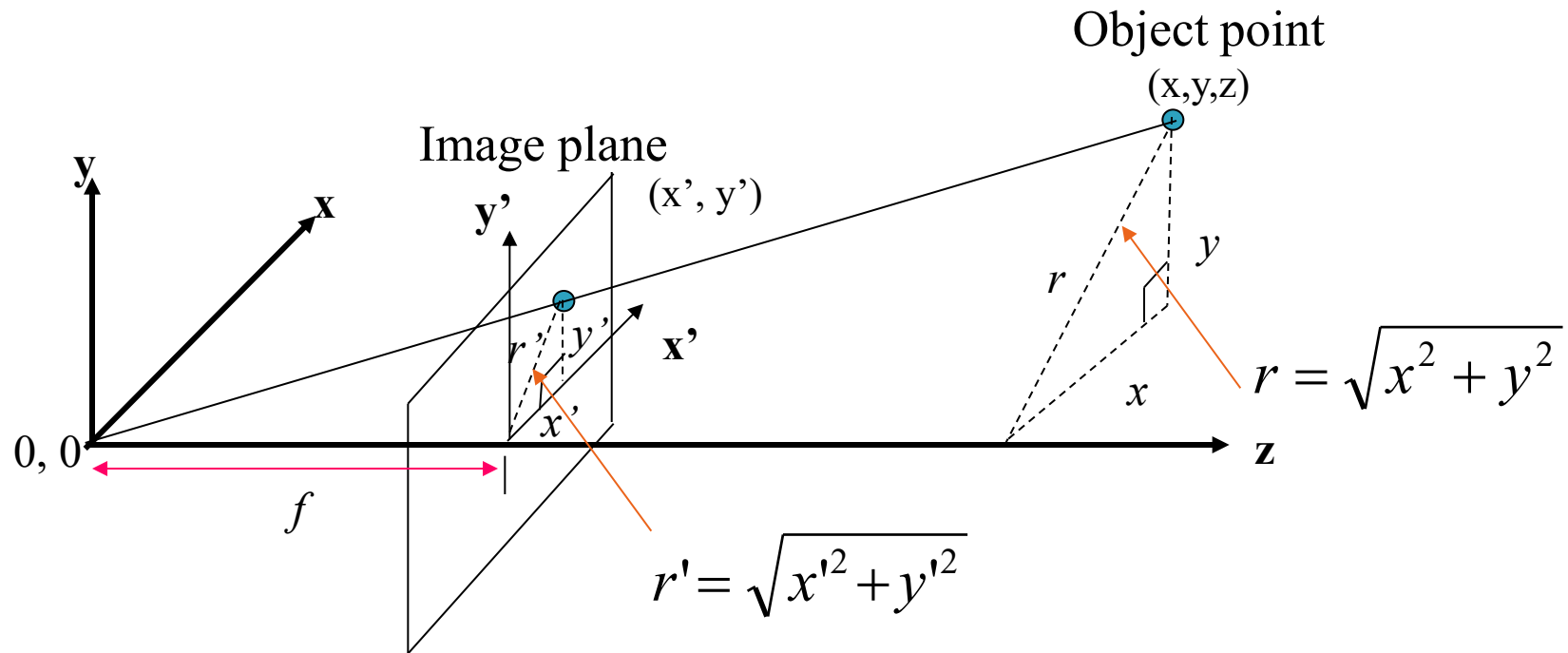
# Image Geometry (cont')

- ▶ The coordinate system for points in the scene is the 3-D space spanned by the unit vectors  $\mathbf{x}$ ,  $\mathbf{y}$ ,  $\mathbf{z}$  that form the axes of the coordinates system.
  - ▶ A point in the scene has coordinates  $(x,y,z)$ .
  - ▶ The  $x$ ,  $y$  coordinates = horizontal and vertical positions of the point in space as seen from the camera (lens).
  - ▶ The  $z$  coordinate = distance from the camera (lens) to the point in space along a line parallel to the  $\mathbf{z}$  axis.
  - ▶ The *line of sight* of a point in the scene is the line that passes through the point of interest and the centre of projection.
- 

# Image Geometry (cont')

- ▶ The *image plane* is parallel to the  $x$  and  $y$  axes of the coordinate system at a distance  $f$  (focal length) from the centre of projection, as shown in the figure.
- ▶ Note that image plane in an actual camera is at a distance  $f$  behind the centre of projection, and the figure is inverted.
- ▶ It is customary to avoid this inversion by assuming that the image plane is in front of the centre of projection as shown in the figure next page.
- ▶ The image plane is spanned by the vectors  $x'$  and  $y'$  to form a 2-D coordinate system for specifying the position of points in the image plane.
- ▶ The position of a point in the image plane is specified by the two coordinates  $x'$  and  $y'$ .
- ▶ Point  $(0, 0)$  in the image plane is its origin.

# Image Geometry (cont')



An illustration showing the *line of sight* that is used to calculate the projected point  $(x', y')$  from the object point  $(x, y, z)$ .

# Perspective projection

- The position  $(x', y')$  in the image plane of a point at position  $(x, y, z)$  in the scene is found by computing the coordinates  $(x', y')$  of the intersection of the line of sight passing through the scene point  $(x, y, z)$  with the image plane.
- The distance of the point  $(x, y, z)$  from the z-axis is, 
$$r = \sqrt{x^2 + y^2}$$
- The distance of the projected point  $(x', y')$  from the origin of the image plane is 
$$r' = \sqrt{x'^2 + y'^2}$$
- The z-axis, the line of sight to point  $(x, y, z)$ , and the line segment of length  $r$  from point  $(x, y, z)$  to the z-axis (perpendicular to the z-axis) form a triangle.
- The z-axis, the line of sight to point  $(x', y')$  in the image plane, and the line segment of length  $r'$  from point  $(x', y')$  to the z-axis (perpendicular to the z-axis) form another triangle.

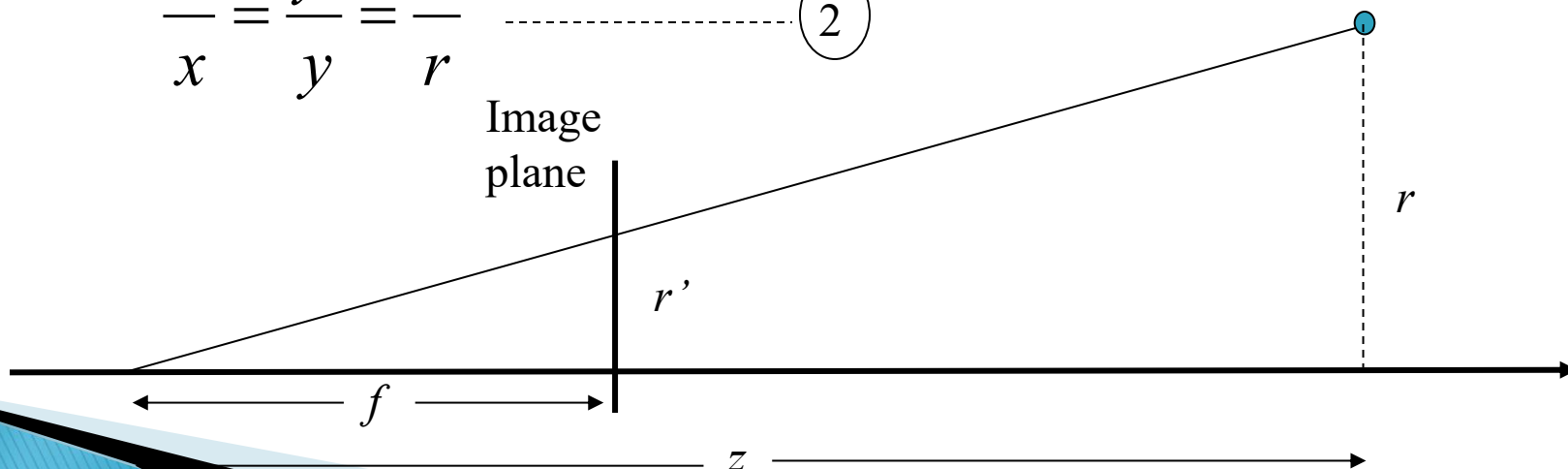
# Perspective projection (cont')

- The two triangles are similar, so the ratios of the corresponding sides of the triangles must be the same:

$$\frac{f}{z} = \frac{r'}{r} \quad \text{-----} \quad (1)$$

- The triangle formed from the  $x$  and  $y$  coordinates and the perpendicular distance  $r$  and the triangle formed from the image plane coordinates  $x'$ ,  $y'$  and the perpendicular distance  $r'$  are also triangles:

$$\frac{x'}{x} = \frac{y'}{y} = \frac{r'}{r} \quad \text{-----} \quad (2)$$



# Perspective projection (cont')

- Combining equations 1 and 2 yields the equation for perspective projection:

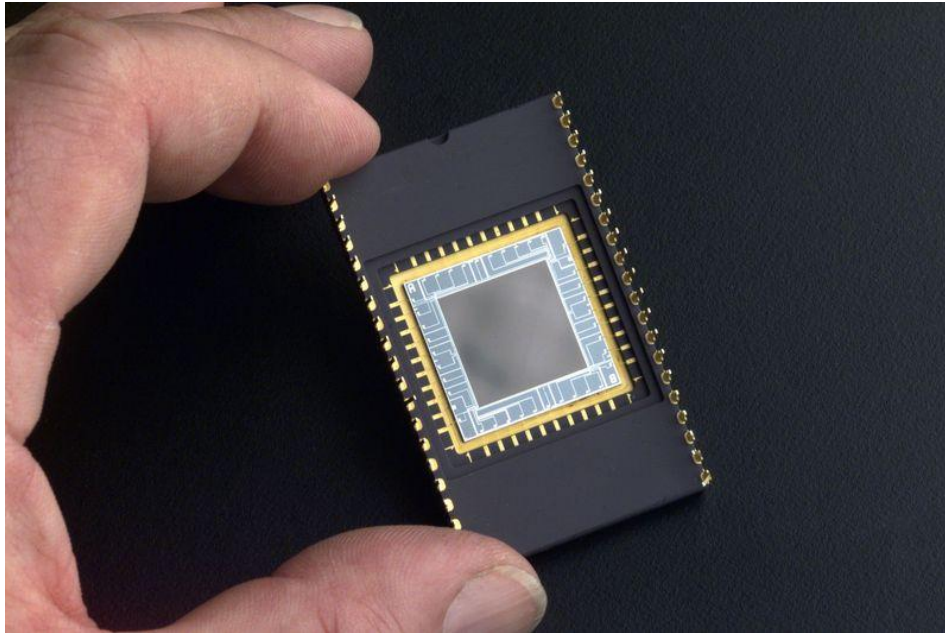
$$\frac{x'}{x} = \frac{f}{z} \quad \text{and} \quad \frac{y'}{y} = \frac{f}{z}$$

- The position of a point  $(x,y,z)$  in the image plane is given by the equations

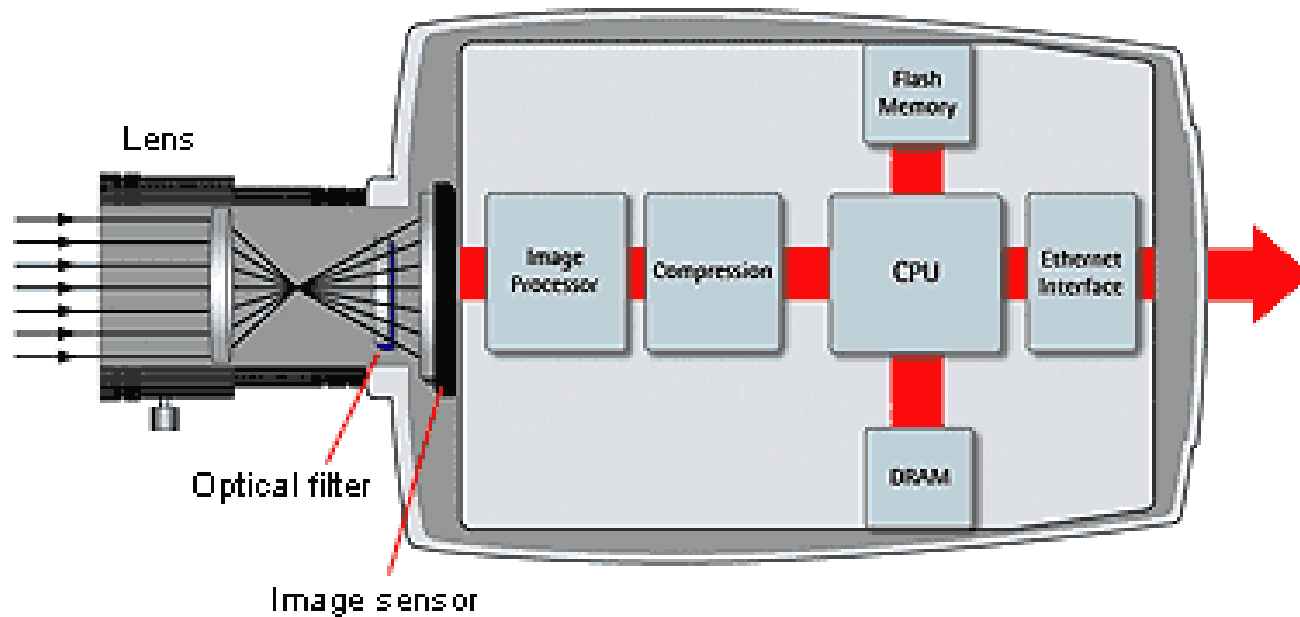
$$x' = \frac{f}{z} x \quad \text{and} \quad y' = \frac{f}{z} y$$

# Imaging Devices: CCD Cameras

- The charge-coupled device (CCD) technology, the most flexible and common input device for machine vision systems.
  - In a CCD camera tiny solid-state cells convert light energy into electrical charge.
  - Each cell converts the light energy it receives into an electrical charge.
  - All cells are first cleared to 0, and then begin to integrate their response to the light energy falling onto them.
  - A shutter may or may not be needed to control the sensing time.
  - The image plane (CCD sensor) acts as a digital memory that can be read row-by-row by a computer input.
- 



# Imaging Devices: CCD Cameras (cont')



Main components of a camera

## Front-End: Capturing Light

- ◆ **Lens:** Focuses incoming light from the scene onto the image sensor. Determines image sharpness, field of view, and depth of field.
- ◆ **Optical Filter:** Filters specific wavelengths (e.g., infrared or UV) before light reaches the sensor. Enhances image quality or enables specialized imaging (e.g., night vision, multispectral).

- ◆ **Image Sensor:** Converts focused light into electrical signals. Can be **CCD** or **CMOS**—each pixel generates a charge based on light intensity.
- ◆ **Image Processor:** Processes raw sensor data into a usable image. Handles tasks like noise reduction, White balance, and Color correction
- ◆ **Compression:** Reduces image file size for storage or transmission – JPEG, PNG, or proprietary codecs.

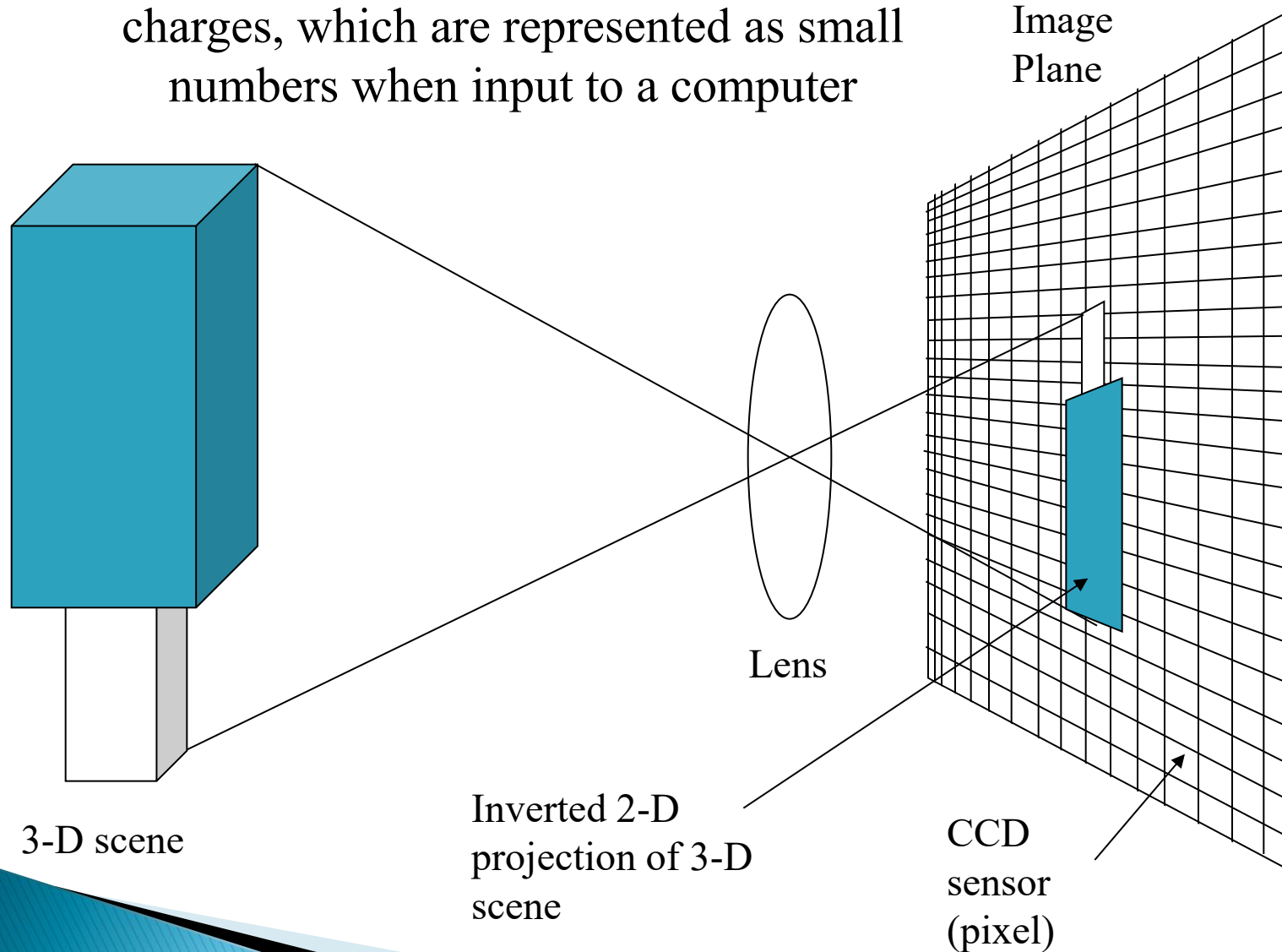
## Back-End:

- ◆ **CPU (Central Processing Unit):** Coordinates all operations: sensor readout, processing, memory access, and data transmission.
- ◆ **DRAM (Dynamic RAM):** Temporary memory used during image processing. Buffers data before compression or storage.
- ◆ **Flash Memory:** Permanent storage for processed images. Can be internal or external (e.g., SD card).
- ◆ **Ethernet Interface:** Enables image data transmission to external systems (e.g., servers, monitoring stations). Useful in industrial, surveillance, or scientific imaging setups.

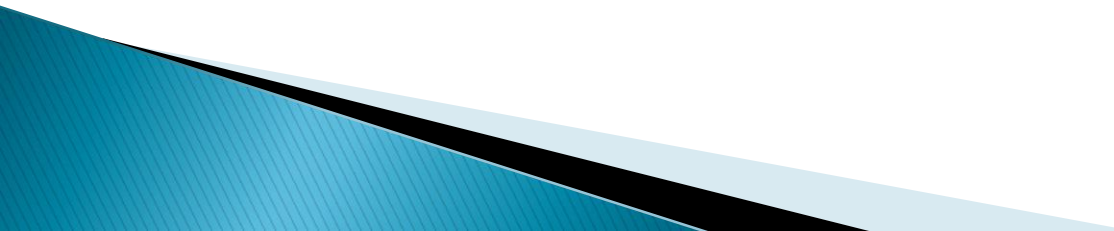
## Data Flow Summary

Light → Lens → Filter → Sensor → Processor → Compression → CPU → DRAM → Flash/Ethernet

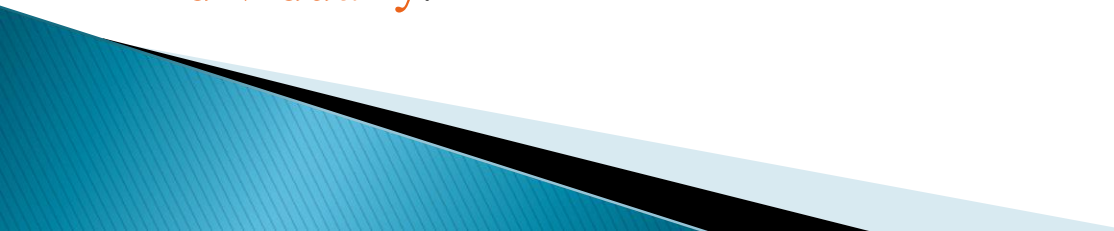
A CCD camera imaging a scene; discrete cells convert light energy into electrical charges, which are represented as small numbers when input to a computer



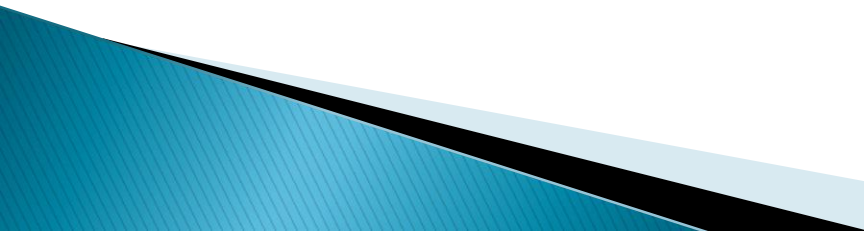
# CCD vs CMOS Cameras

- Both CCD (charge-coupled device) and CMOS (complimentary metal-oxide semiconductor) image sensors start at the same point -- they have to convert light into electrons.
  - One simplified way to think about the sensor used in a digital camera (or camcorder) is to think of it as having a 2-D array of thousands or millions of tiny solar cells, each of which transforms the light from one small portion of the image into electrons.
  - Both CCD and CMOS devices perform this task using a variety of technologies.
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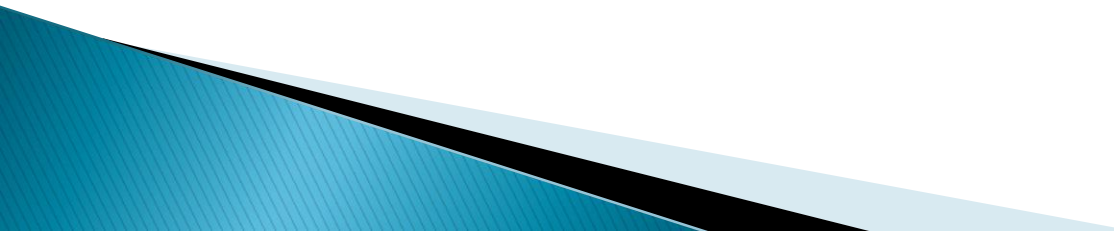
# CCD vs CMOS Cameras (cont')

- Next step is to read the value (accumulated charge) of each cell in the image.
  - In a CCD device, the charge is actually transported across the chip and read at one corner of the array.
  - An analog-to-digital converter turns each pixel's value into a digital value.
  - In most CMOS devices, there are several transistors at each pixel that amplify and move the charge using more traditional wires.
  - The CMOS approach is more flexible because each pixel can be read individually.
- 

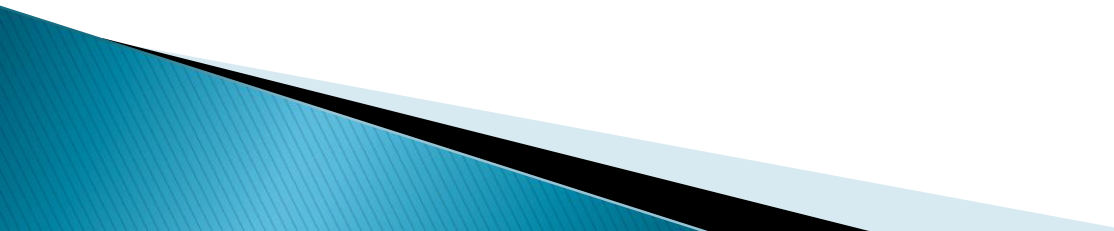
# CCD vs CMOS Cameras (cont')

- CCDs use a **special** manufacturing process to create the ability to transport charge across the chip without distortion.
  - This process leads to very high-quality sensors in terms of **fidelity** and **light sensitivity**.
  - CMOS chips, on the other hand, use **traditional** manufacturing processes to create the chip -- the same processes used to make most microprocessors.
  - Because of the manufacturing differences, there have been some noticeable differences between CCD and CMOS sensors.
- 

# CCD vs CMOS Cameras (cont')

- **CCD sensors**, as mentioned above, **create high-quality, low-noise images**. **CMOS sensors**, traditionally, are **more susceptible to noise**.
  - Because each pixel on a CMOS sensor has several transistors located next to it, the **light sensitivity of a CMOS chip tends to be lower**. Many of the photons hitting the chip hit the transistors instead of the photodiode.
  - **CMOS traditionally consumes little power**. Implementing a sensor in CMOS yields a low-power sensor.
  - **CCDs use a process that consumes lots of power**. CCDs consume as much as 100 times more power than an equivalent CMOS sensor.
  - CMOS chips can be fabricated on just about any standard silicon production line, so they tend to be extremely **inexpensive compared to CCD sensors**.
  - CCD sensors have been mass produced for a longer period of time, so they are more mature. They tend to have **higher quality and more pixels**.
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# CCD vs CMOS Cameras (cont')

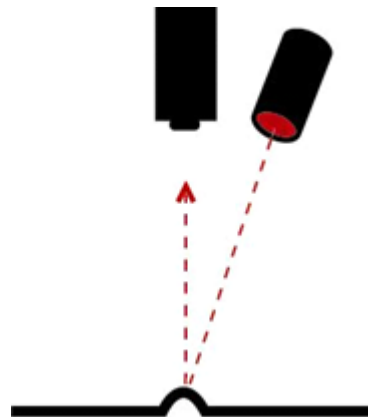
- Based on these differences, we can see that CCDs tend to be used in cameras that focus on **high-quality images** with **lots of pixels** and **excellent light sensitivity**. CMOS sensors traditionally have **lower quality, lower resolution** and **lower sensitivity**.
  - CMOS sensors are just now improving to the point where they reach near parity with CCD devices in some applications. CMOS cameras are usually **less expensive** and have **great battery life**.
- 

# Lighting Systems

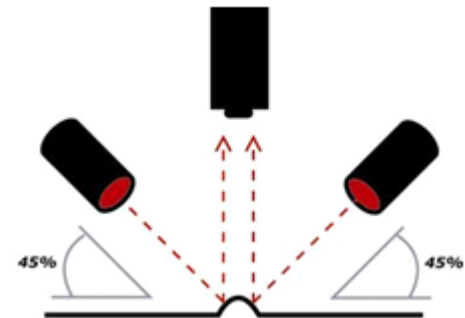
- There are two basic lighting set-ups,
  - front lighting
  - back lighting.
- Depends on the inspection required:
  - front lighting is good for **highlighting surface items**, such as print, etched numbers, etc
  - back lighting provides **better contrast**, for example, gauging and positioning tasks.



Back lighting



Directional bright field



Directional dark field

Front lighting

## Directional Lighting

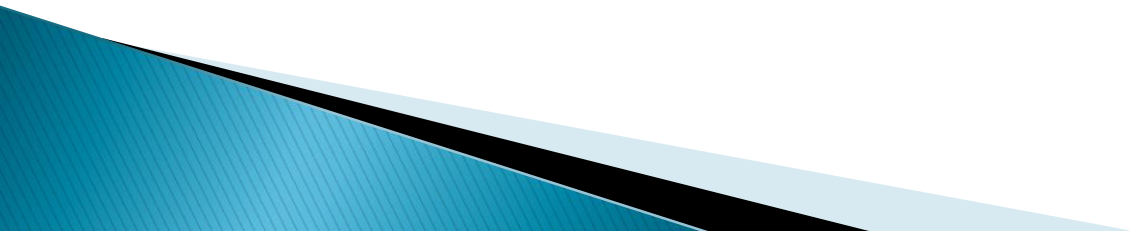


## Diffused Lighting



# Lighting Systems (cont')

- Directional lighting is great for picking out surface effects, just as human tilts an object to view it from different angles to help identify surface structures but shadows may be a problem.
- Diffused lighting helps eliminate shadows and reflections making it better for positional checking.



# Lighting Systems (cont')

- Another way to eliminate reflections - and improve contrast on transparent pieces is to polarize the light source and use a polarizing filter on the camera (polarized back lighting can even render surface tensions visible).
- Ring lights provide intense shadow-free lighting along the optical axis and are often combined with polarizing filters to remove troublesome reflections.
- It is also worth remembering that infra-red and ultra-violet can be used - with a suitable camera - as light sources to achieve the required illumination.

Further reading:

<http://www.machinevisiononline.org/public/articles/archivedetails.cfm?id=1430>

<http://www.machinevisiononline.org/public/articles/articlesdetails.cfm?id=131>

# Lighting Systems (cont')



# A Simple Image Model

- ▶ Image  $\rightarrow$  2D light-intensity function,  $f(x,y)$  where the value or amplitude of  $f$  at spatial coordinates  $(x,y)$  gives the brightness of the image at that point.
- ▶ As light is a form of energy,  $f(x,y)$  must be nonzero and finite:  $0 < f(x,y) < \infty$
- ▶ The basic nature of  $f(x,y)$  may be characterized by 2 components:
  - Amount of source light incident on the scene being viewed (**illumination**)
  - Amount of light reflected by the objects in the scene (**reflectance**)

# A Simple Image Model (cont')

- ▶ Let illumination and reflectance components be denoted by  $i(x,y)$  and  $r(x,y)$  are combined as a product to form  $f(x,y)$ :

$$f(x,y) = i(x,y)r(x,y)$$

where  $0 < i(x,y) < \infty$   
and  $0 < r(x,y) < 1$

- ▶ The nature of  $i(x,y)$  is determined by the **light source** and  $r(x,y)$  is determined by the **characteristics of the objects** in the scene.

# A Simple Image Model (cont')

- ▶ Average values of illumination and reflectance:

|                   |                          |
|-------------------|--------------------------|
| Clear day         | 90,000 lm/m <sup>2</sup> |
| Cloudy day        | 10,000 lm/m <sup>2</sup> |
| Full moon         | 0.1 lm/m <sup>2</sup>    |
| Commercial office | 1000 lm/m <sup>2</sup>   |

|                       |             |
|-----------------------|-------------|
| Black velvet          | <b>0.01</b> |
| Stainless steel       | <b>0.65</b> |
| Flat-white wall paint | <b>0.80</b> |
| Silver-plated metal   | <b>0.90</b> |
| Snow                  | <b>0.93</b> |

# A Simple Image Model (cont')

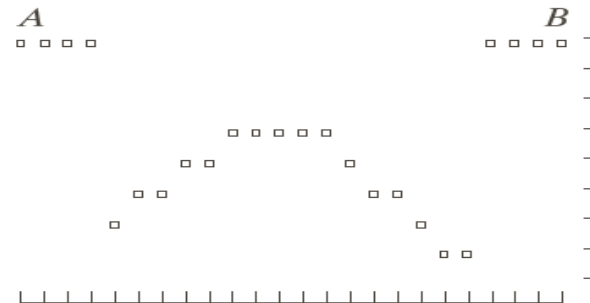
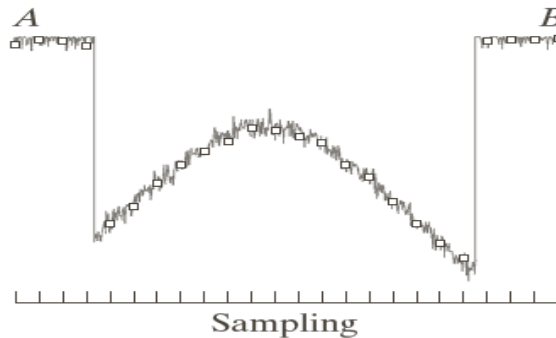
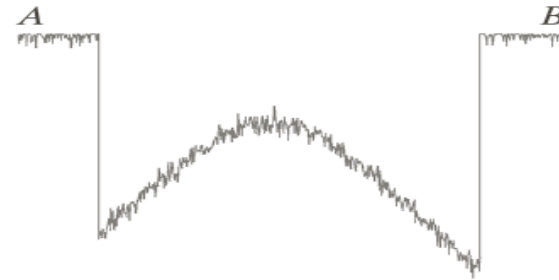
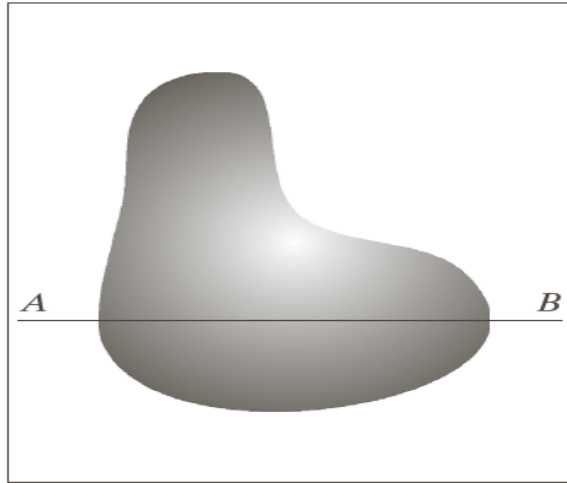
## Test yourself:

- ▶ Intensity of a monochrome image at a coordinate  $(x_0, y_0)$  is called the gray level of the image at that point. This gray level can be in the range of  $[L_{min}, L_{max}]$  where  $L_{min} = i_{min} r_{min}$  and  $L_{max} = i_{max} r_{max}$ . Using the preceding average office illumination and range of reflectance value, what will be the typical limits  $[L_{min}, L_{max}]$  for indoor values in the absence of additional illumination?

# Sampling and Quantization

- ▶ Continuous functions cannot be represented exactly in a digital computer
- ▶ Interface between the optical system that projects a scene onto the image plane and the computer must sample the image at a finite number of points and represent (quantize) each sample within the finite word size of the computer/machine.
- ▶ This process is known as *sampling* and *quantization* which transform analog images to digital images.
- ▶ Each image sample is called a **pixel** (short for picture element).
- ▶ Some literatures also call it *image element*, *picture element* and *pel*.

# Sampling and Quantization



**Sampling** – Digitizing the coordinate values

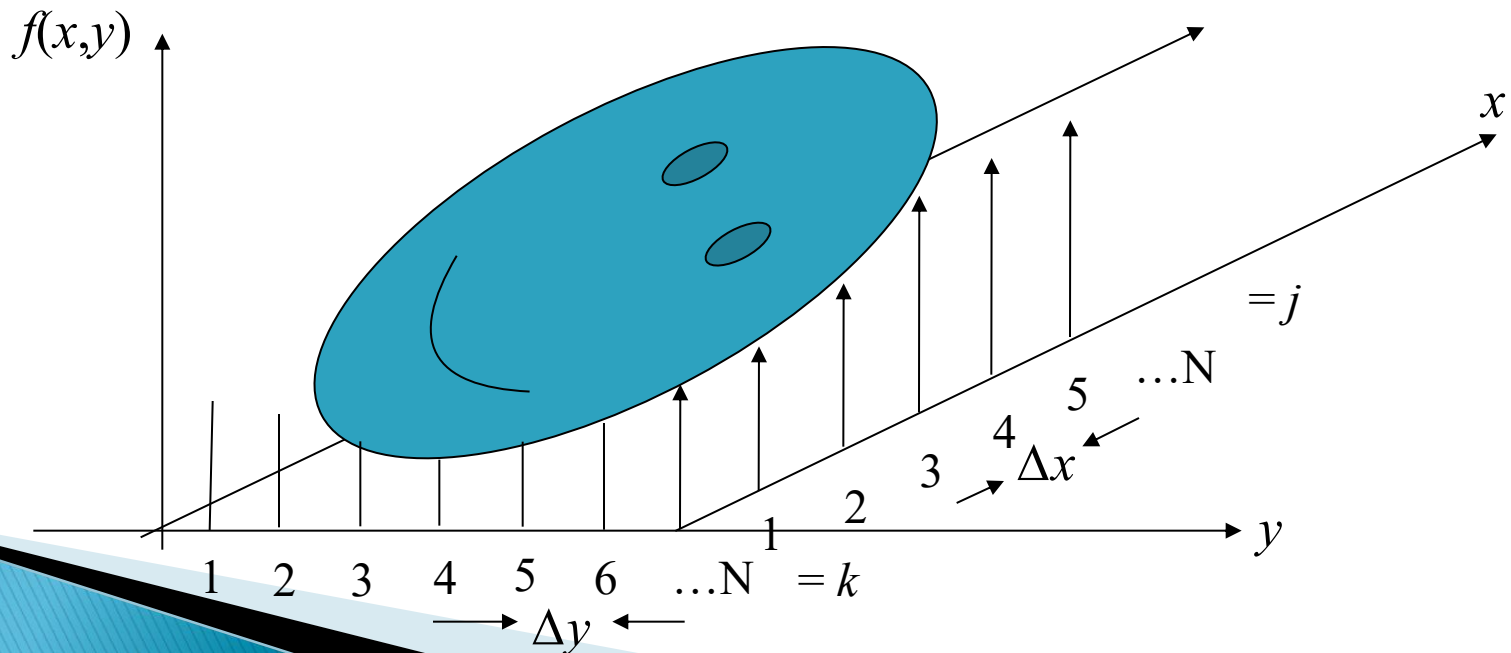
**Quantization** – Digitizing the amplitude values

# Sampling

- ▶ Let  $f(x,y)$  be a continuous-valued intensity image and let  $s(x,y)$  be a 2-dimensional sampling function of the form

$$s(x,y) = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \delta(x - j\Delta x, y - k\Delta y)$$

- ▶ The 2-dimensional sampling function is an infinite array of Dirac delta function as shown in the figure below:



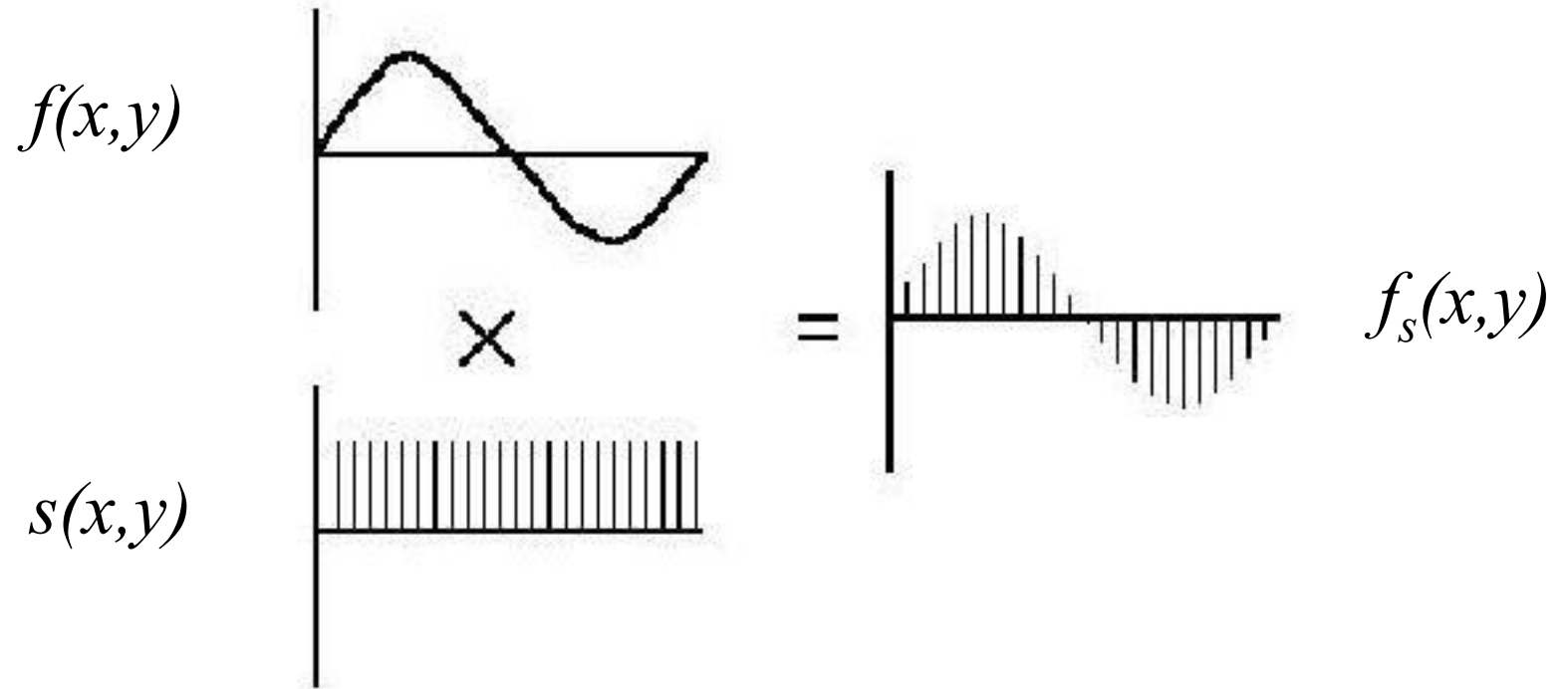
# Sampling (cont')

- ▶ The sampling function also known as the **comb function**, is arranged in a regular grid of spacing  $\Delta x$  and  $\Delta y$  along the  $X$  and  $Y$  axes respectively. The sampled image may be represented as

$$f_s(x, y) = f(x, y)s(x, y)$$
$$\sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} f(j\Delta x, k\Delta y) \cdot \delta(x - j\Delta x, y - k\Delta y)$$

- ▶ The sampled image  $f_s(x, y)$  is an array of image intensity values at the sample points  $(j\Delta x, k\Delta y)$  in a regular 2-dimensional grid.

# Sampling (cont')

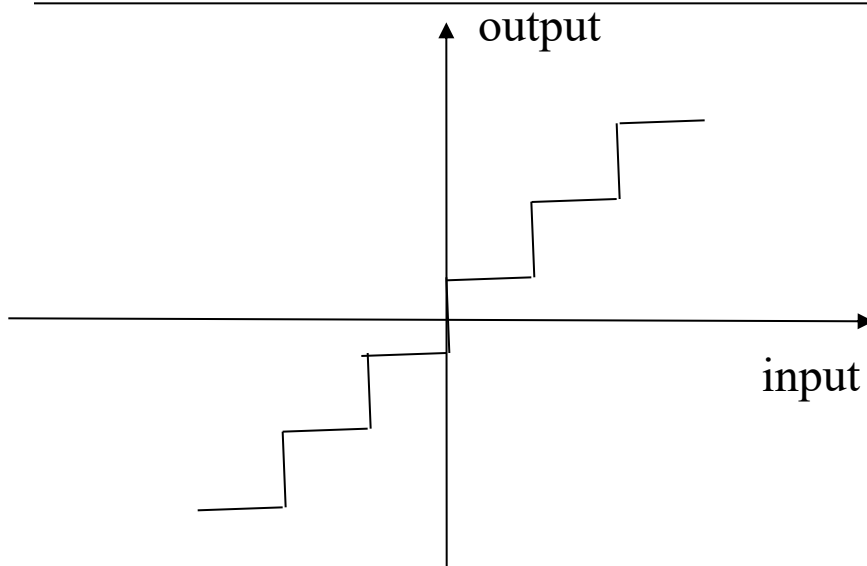


# Quantization

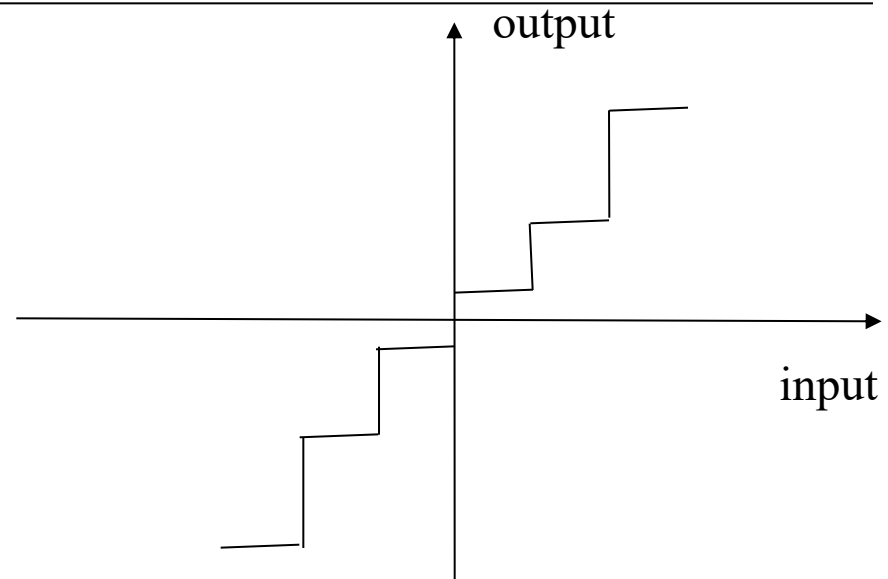
- ▶ Conversion of the sampled analog pixel intensities to discrete valued integer numbers is the process of *quantization*.
- ▶ Quantization assigns a single value to each sample in such a way that the image reconstructed from the quantized values are good in quality with small quantization error.
- ▶ Some interesting questions:
  - How many quantized levels are sufficient to represent each sample?
  - How do we choose the quantization levels?
- ▶ As number of quantization levels increases, the original image is approximated in a better way and less quantization error.
- ▶ When the quantization levels are chosen equally spaced at equal interval, it is known as *uniform quantization*.

# Quantization (cont')

- ▶ When the sample intensity values are equally likely to occur at different intervals, **uniform quantization** is always preferred.
- ▶ However, in some situations, the image samples assume values in a small range quite frequently and other values infrequently.
- ▶ In such situations, it's preferable to use *nonuniform quantization*.

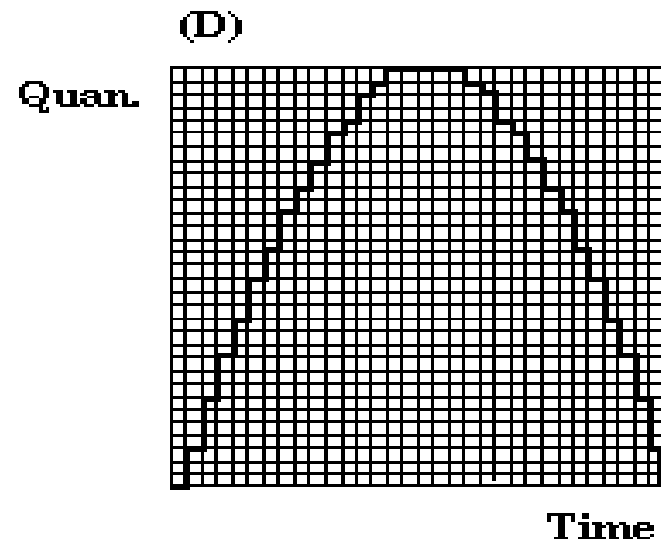
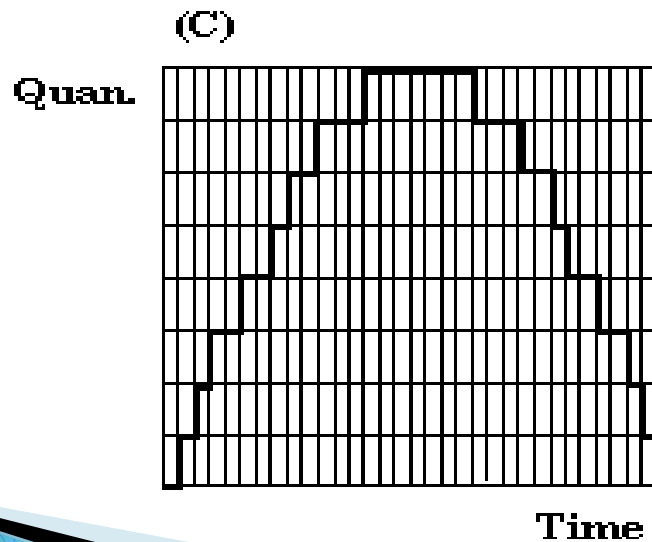
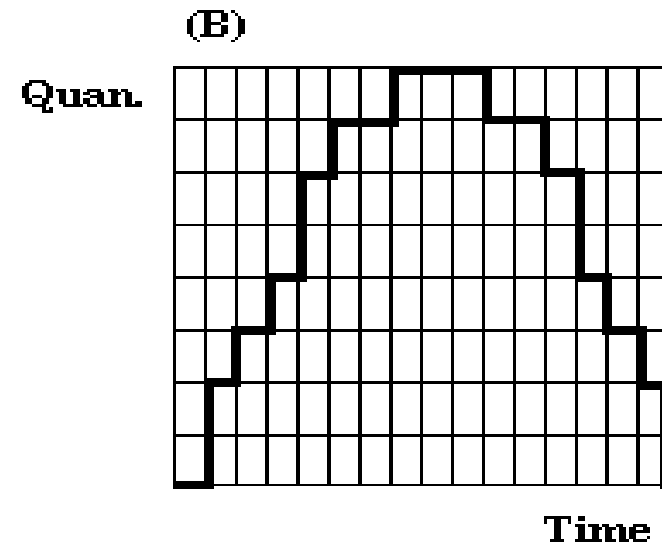
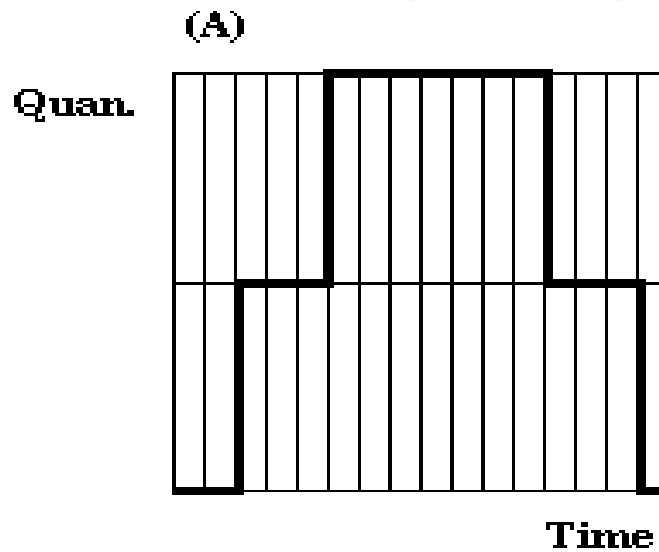


2-D uniform quantization



2-D nonuniform quantization

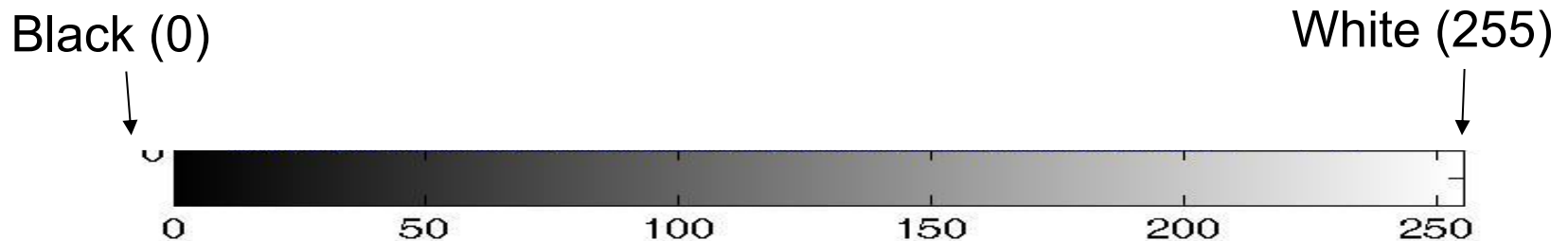
# Quantization (cont')

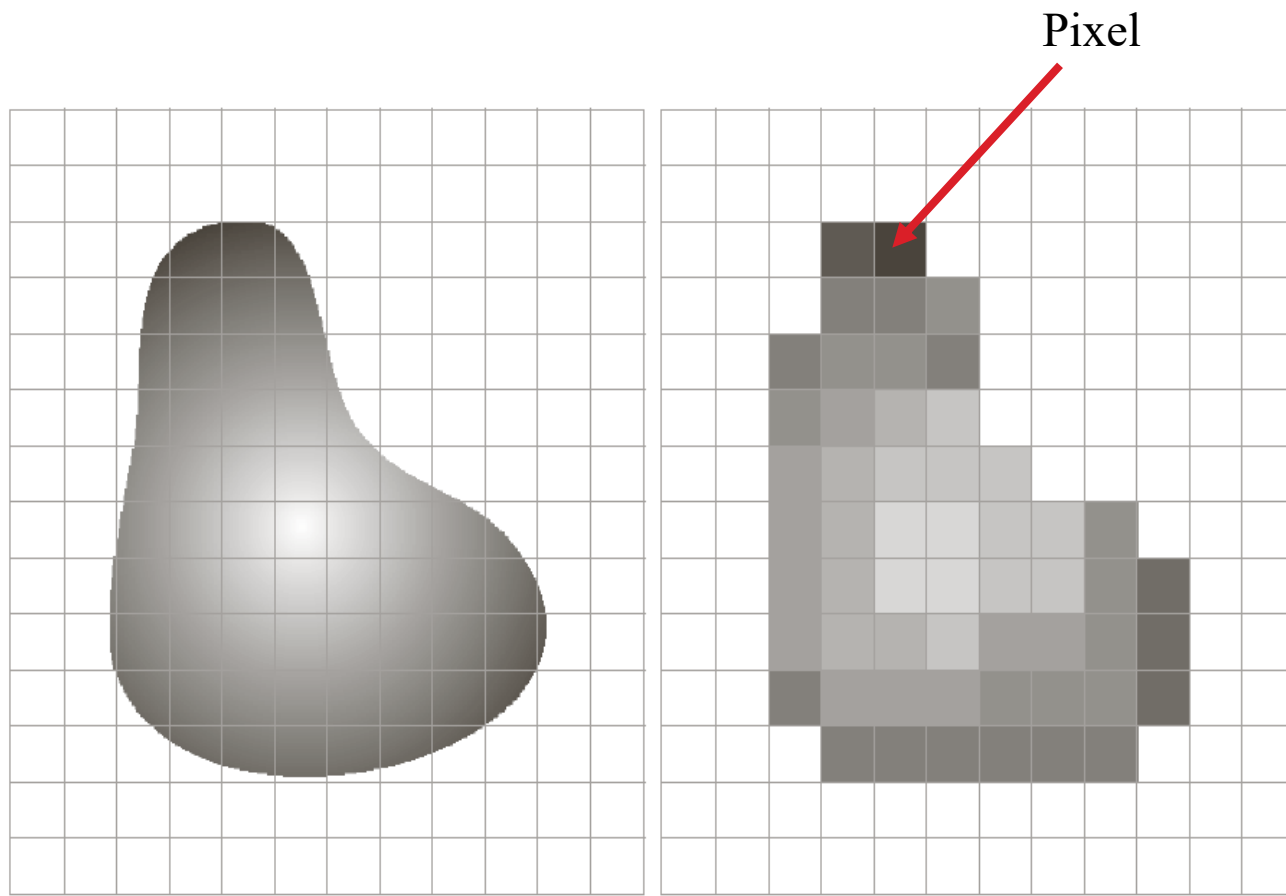


# Quantization (cont')

- ▶ Throughout this course, we assume that images are sampled on a rectangular grid of squares so that the horizontal and vertical distances between pixels are same throughout the digital image, as in most cases.
- ▶ A pixel may be represented as an unsigned 8-bit integer in the range  $[0,255]$ .
- ▶  $0 \rightarrow$  black,  $255 \rightarrow$  white, shades of gray distributed over the middle values.

Gray levels for an unsigned 8-bit integer image





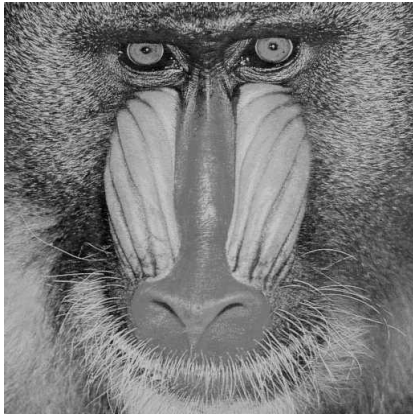
Continuous image

Result of sampling  
and quantization

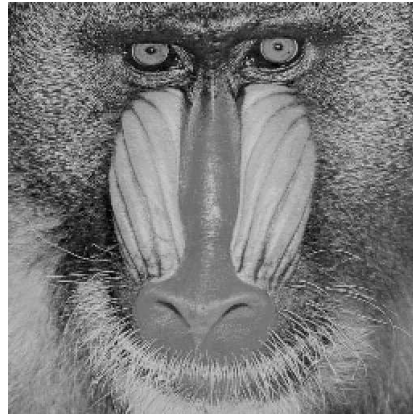


# Quantization (cont')

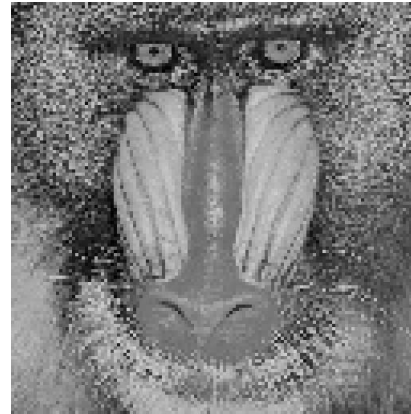
512 x 512



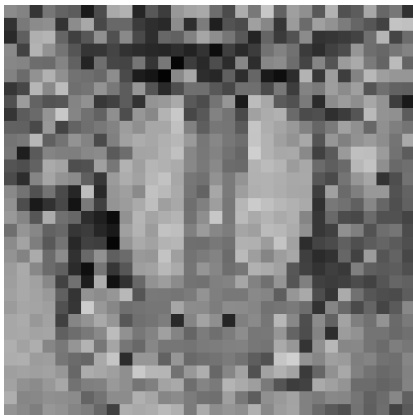
256 x 256



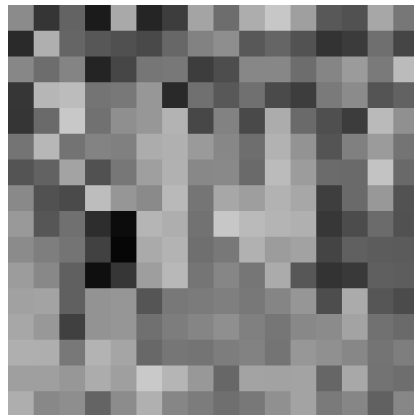
128 x 128



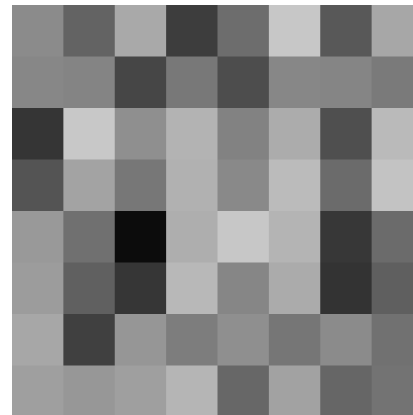
64 x 64



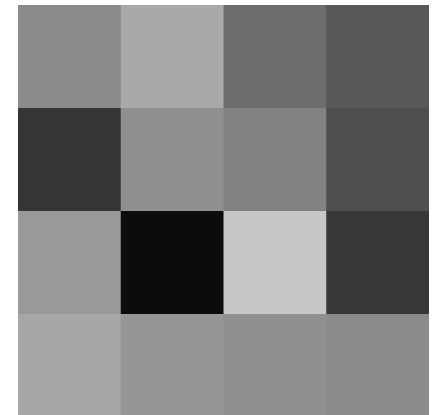
32 x 32



16 x 16



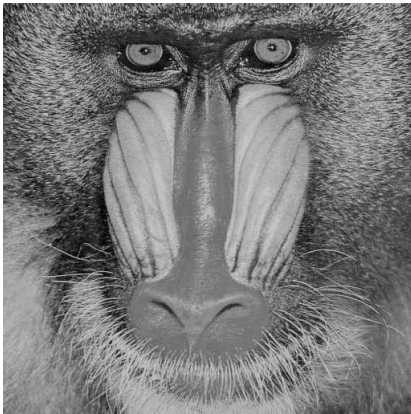
8 x 8



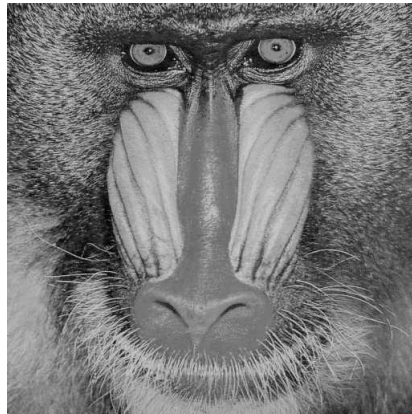
4 x 4

# Quantization (cont')

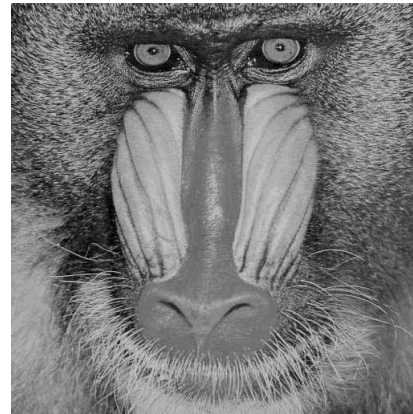
256 graylevels:  
8 bits



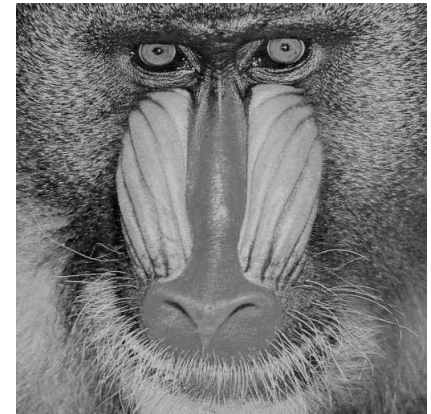
128 graylevels :  
7 bits



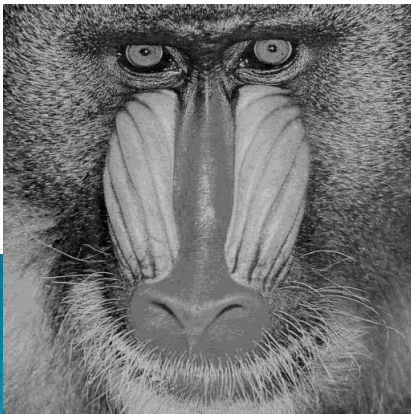
64 graylevels:  
6 bits



32 graylevels:  
5 bits



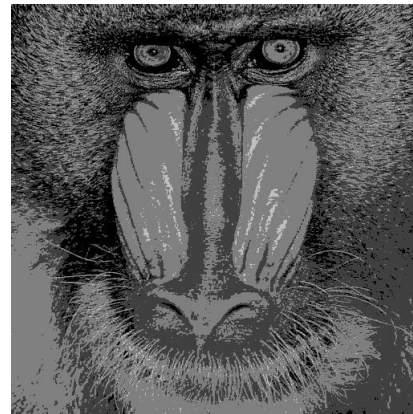
16 graylevels:  
4 bits



8 graylevels :  
3 bits



4 graylevels:  
2 bits



2 graylevels:  
1 bit (binary)



- ▶ **Sampling rate** determines how many pixels the digital image has (resolution) -> discretization in space
- ▶ **Quantization levels** determine how many intensity levels will be used -> discretization in intensity
- ▶ In most machine vision applications, the sampling rate and the quantization levels are predetermined.
- ▶ However, in many applications it is important to know the effects of sampling and quantization in order to retain the important information in images.

# Image Sampling and Quantization (cont')

- ▶ Effect of sampling → checkerboard effect

1024x1024



512x512



256x256



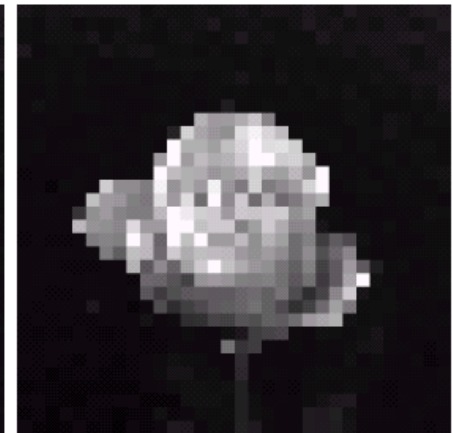
128x128



64x64

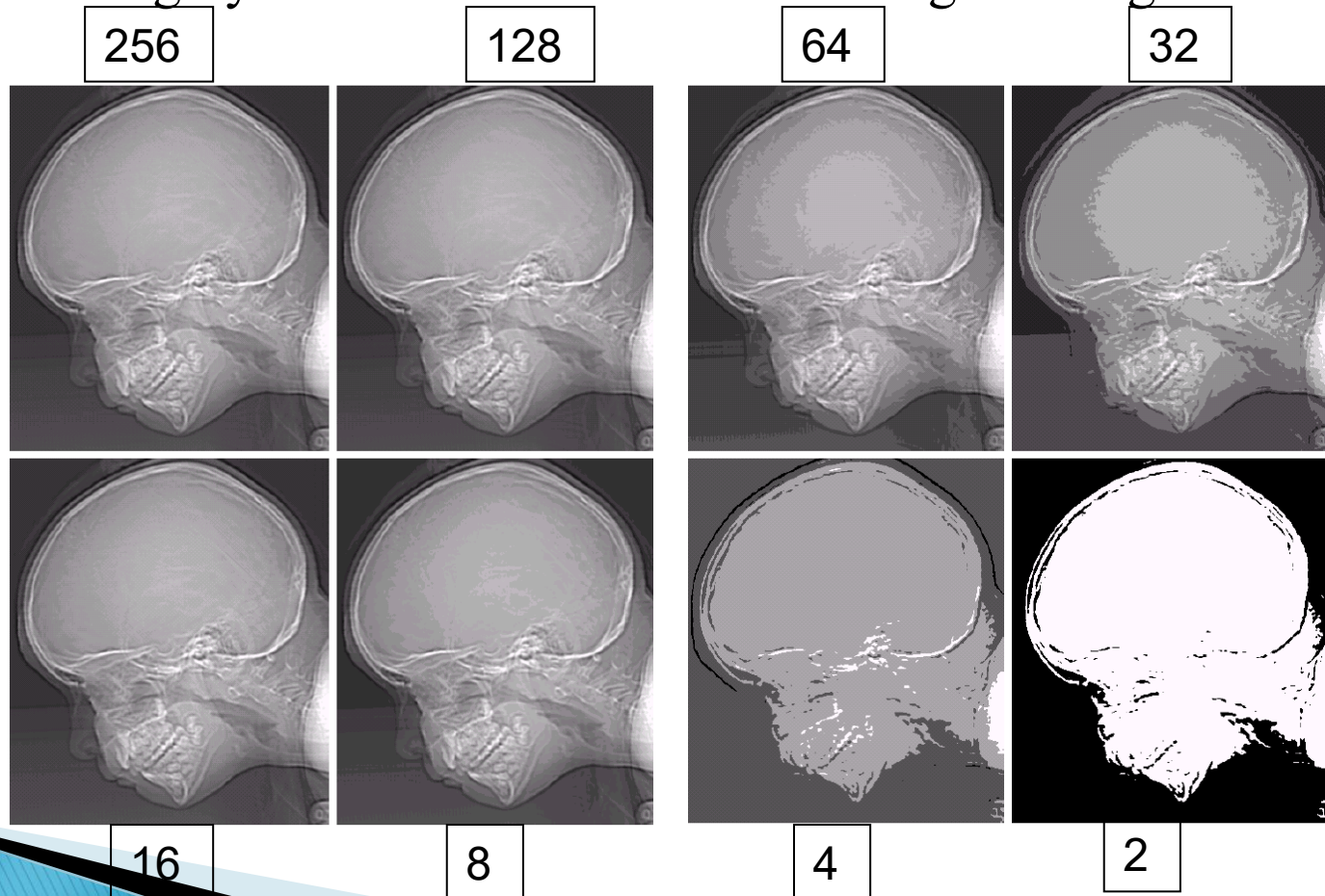


32x32



# Image Sampling and Quantization (cont')

- ▶ Effect of quantization → false contouring → insufficient number of gray levels in smooth areas of a digital image

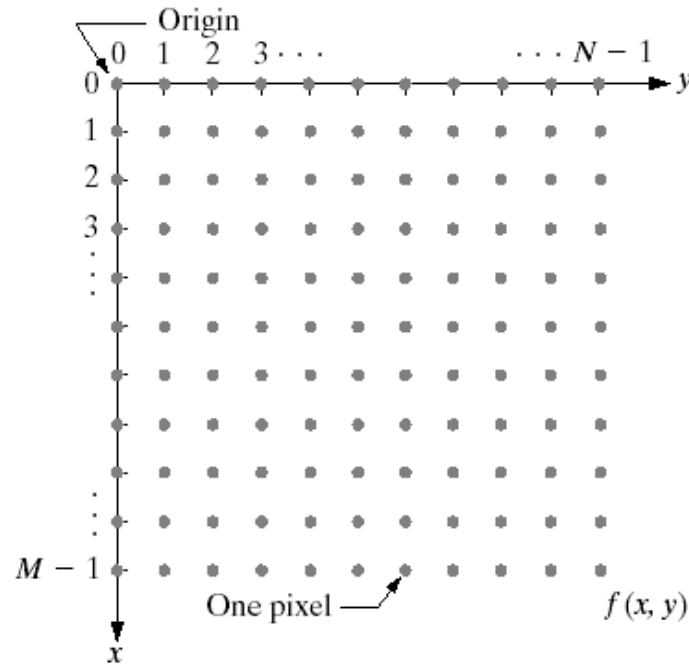


# Digital Image Representation

- ▶ The term *monochrome image*, or simply *image* refers to a 2-dimensional light intensity function  $f(x,y)$ , where  $x$  and  $y$  denote **spatial coordinates** and the value of  $f$  at any point  $(x,y)$  is proportional to the **brightness** (or **gray level**) of the image at that point.
- ▶ The figure next page illustrates the axis convention used throughout this course.
- ▶ A *digital image* is an image  $f(x,y)$  that has been discretized both in **spatial coordinates** and **brightness/intensity**.
- ▶ A digital image can be considered a matrix whose row and column indices identify a point in the image and the corresponding matrix element value identify an intensity in the image at that point.

# Digital Image Representation (cont')

- Assume that an image  $f(x,y)$  is sampled so that the resulting digital image has  $M$  rows and  $N$  columns.
- The coordinate convention for representing digital image is:



$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

# Digital Image Representation (cont')

- ▶ Digitization process requires decisions about values for  $M$ ,  $N$ , and number of,  $L$ , discrete gray level allowed for each pixel.  $L = 2^k$  (integer power of 2)
- ▶ Discrete gray levels interval  $[0, L-1]$  → dynamic range
- ▶ Images with gray levels span significant portion of gray scales refer as having high dynamic range → high contrast
- ▶ Low dynamic range → dull, washed out gray look.

# Digital Image Representation (cont')

- ▶ Number of bits,  $b$ , required to store a digital image is:

$$b = M \times N \times k$$

- *Example:*

A 3 bits image has a dimension of 128 (M) x 96 (N). What number of bits required to store the digitized image?

- *Answer:*

$k = 3$ ; Number of bits,  $b = 128 \times 96 \times 3 = 36864$  bits = 4608 bytes.

- ▶ **Spatial resolution** → smallest discernible line pairs per unit distance.
- ▶ **Gray level resolution** → smallest discernible change in gray level.
- ▶ Common notation: an  $L$ -level digital image of size  $M \times N$  as having **spatial resolution** of  $M \times N$  pixels and a **gray level resolution** of  $L$  levels.